

# Akhil Reddy Vangala

## Data Analyst

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### SUMMARY

**Data Analyst with 5+ years of experience** in assisting with the design, development, and maintenance of scalable ETL pipelines and building machine learning models to support data-driven decision-making in financial and healthcare domains. Proficient in performing exploratory data analysis (EDA), statistical modeling, and anomaly detection on large, complex datasets using Python (Pandas, NumPy, Scikit-learn, PySpark), R, and SQL. Skilled in handling big data environments using Apache Spark, Hadoop, and Kafka, with practical expertise across major cloud platforms including AWS, Azure, and GCP. Adept at transforming raw data into actionable insights through data wrangling, feature engineering, and real-time analytics solutions. Experienced in creating dynamic dashboards and regulatory reports using Power BI, Tableau, and Python Dash. Hands-on with NLP, Generative AI (LLMs), fraud detection models, and MLOps tools such as MLflow and Azure ML. Strong understanding of SDLC, Agile frameworks, data governance, and model interpretability techniques including SHAP.

### SKILLS

**Methodologies:** SDLC, Agile, Waterfall

**Language:** SQL, Python (NumPy, Pandas, PySpark, SciPy, Scikit-Learn, TensorFlow, PyTorch, Keras), R

**Visualization:** Tableau, Power BI, Matplotlib, Seaborn, ggplot2

**Tools:** Microsoft Excel, Spark, Advanced Excel, SAS JMP, Docker, Azure, GCP, Visio, SAS Enterprise Miner, RASA, Neo4j, SSIS, MySQL, SQL Server, Oracle, Postgres, Big Query, Databricks, AWS, Jira

**Techniques:** Statistics, Data Visualization, ETL Process, Machine Learning, Predictive Modeling, Data Modeling, Probability, Optimization, Pattern Recognition, Linear Regression, Logistic Regression, Decision Trees, Random Forests, Correlations, Hypothesis Testing, Multivariate Analysis, Clustering, Natural Language Understanding, Natural Language Processing, Project Management

**Operating System:** Windows, Linux

### WORK EXPERIENCE

**JPMorgan Chase & Co., USA | Data Analyst**

**June 2024 - Current**

- Designed and maintained scalable **ETL pipelines** using **Apache Spark** and **SQL Server**, processing over **10M+ daily transactions** for **fraud** and **AML analytics**.
- Developed **anomaly detection models** using **Python (Scikit-learn, PyOD)** and **statistical methods** to flag suspicious transaction patterns in **real-time**.
- Utilized **unsupervised machine learning** techniques such as **K-Means clustering** and **Isolation Forests** to enhance **risk segmentation** and uncover hidden **fraud trends**.
- Built interactive dashboards with **Tableau** and **Python Dash** to visualize alerts, transaction clusters, and **compliance KPIs** for **regulatory reporting**.
- Integrated **Big Data** frameworks like **Hive on Hadoop** to manage and query **structured** and **semi-structured transaction logs** across **distributed environments**.
- Deployed **machine learning models** to production using **MLflow** and **AWS SageMaker**, enabling **continuous fraud detection** and **model monitoring**.
- Automated **data validation** and **cleansing** processes using **Python (Pandas, NumPy)** and **SQL**, improving **data quality** and reducing **false positive alerts** by **30%**.
- Collaborated with **Compliance** and **Risk teams** to define **fraud risk thresholds**, integrate **model insights**, and support **SAR (Suspicious Activity Reports)** filing.
- Implemented **Generative AI** techniques (**LLMs**) to **auto-summarize** flagged transactions and generate **regulatory narratives**, accelerating **audit readiness**.
- Leveraged **AWS services** such as **S3, Lambda, and Glue** for **secure storage, orchestration, and real-time processing** of **large-scale transactional datasets**.
- Performed **feature engineering** and **model tuning** using **GridSearchCV** and **SHAP** to improve model **precision, recall, and explainability**.
- Utilized **Git, Jupyter Notebooks, and AWS CodePipeline** for **CI/CD workflows** and effective **version control**.
- Conducted **A/B testing** and **impact analysis** to assess **model performance** improvements, resulting in a **22% uplift** in **fraud capture rate** with minimal operational overhead.

- Designed and implemented **real-time data pipelines** using **Apache Kafka** and **Apache Spark Streaming** to ingest, process, and monitor **laboratory test results** from **Electronic Health Record (EHR)** systems.
- Developed scalable **ETL workflows** using **PySpark**, **SQL**, and **Apache Hive** to cleanse, normalize, and structure **raw clinical data** for downstream **alert generation**.
- Built **anomaly detection rules** using **statistical thresholds** and **machine learning models** (e.g., **Isolation Forest**) to flag **outlier lab results** impacting **patient safety**.
- Integrated **clinical alerting logic** into **Spark Structured Streaming** workflows for **real-time notifications** to clinicians via **secure hospital portals**.
- Created **interactive dashboards** using **Power BI** and **Tableau** to track **lab value distributions**, **flagged anomalies**, and **alert trends** across departments.
- Collaborated with **physicians**, **lab technicians**, and **compliance teams** to define **alert thresholds** and **critical value cutoffs** per **CMS** and **HEDIS** regulatory standards.
- Optimized **Kafka topic design**, **message serialization** with **Avro/JSON**, and **partitioning strategies** for **low-latency clinical data flow**.
- Utilized **Azure Synapse Analytics** and **Azure Data Lake** to store **streaming data snapshots** for **historical analysis** and **compliance audits**.
- Deployed **real-time monitoring** tools like **Grafana** and **Prometheus** to visualize **alert pipeline health**, **latency**, and **error rates**.
- Applied **Python (Pandas, NumPy)** for **post-processing**, **data validation**, and **exploratory data analysis (EDA)** on **lab anomalies**.
- Implemented **unit testing**, **data quality checks**, and **schema validation** using **Great Expectations** and **pytest** frameworks.
- Version-controlled the **codebase** using **Git**, and orchestrated **pipeline releases** through **Azure DevOps CI/CD pipelines** for **streamlined deployment**.
- Incorporated **LLM-based summarization models** (e.g., **OpenAI GPT**, **BioGPT**) to generate **clinical notes** and **narrative summaries** for flagged results.
- Built and deployed **machine learning models** with **MLflow** to predict **abnormal lab trends** based on **patient demographics** and **prior tests**.
- Conducted **root cause analysis** and **A/B testing** on **alert outcomes** to continuously refine **detection logic** and reduce **alert fatigue** by **28%**.

## **EDUCATION**

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### **Master's in Data Science**

University of New Haven, West Haven, Connecticut